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# Financial Derivatives

Session 13 & 14: Financial Derivatives

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# What is a derivative?

A contract whose value **derives from an underlying**: a stock, index (Nifty 50), commodity, currency (USD/INR) or interest rate.

## Engineering analogy

A derivative is a function `f(S)`: input = underlying price, output = payout. You hold the rule, not the asset, like a strain gauge reporting on a bridge.

## Three purposes

- **Hedging**: transfer risk
- **Speculation**: view with leverage
- **Arbitrage**: exploit price gaps

## Who uses them

- Exporters/importers (currency)
- Airlines (fuel)
- Farmers (crops)
- Funds and banks (portfolios)

# Key vocabulary

| Term | Meaning |
|---|---|
| Underlying | Asset the contract is based on |
| Strike (K) | Agreed price at which the deal happens |
| Premium | Price paid by option buyer for the right |
| Expiry | Date the contract ends |
| Lot size | Fixed quantity per contract |
| Long / Short | Buyer / seller of the contract |
| Margin | Deposit that covers potential losses |
| Exchange vs OTC | Standardised and cleared vs privately negotiated |

# Underlying assets: what can a derivative track?

| Class | Example | Typical use | Settlement |
|---|---|---|---|
| Equity | Reliance, Infosys shares | Hedge or income on holdings | Cash (shares in some cases) |
| Index | Nifty 50, Bank Nifty | Portfolio hedge, market view | Cash |
| Currency | USD/INR | Exporters and importers hedge | Cash in INR |
| Commodity | Gold, crude oil | Producers and buyers lock prices | Cash or physical |
| Interest rate | Bond yields | Banks manage borrowing cost | Cash |

**Cash vs physical settlement:** index options pay out the rupee difference only; nobody delivers 50 stocks.

# Exchange trading: clearing house and margin

## Novation

Once matched, the **clearing corporation** becomes the buyer to every seller and the seller to every buyer. You never face the other trader directly, so default risk is tiny.

## Margin

A good-faith deposit. **Initial margin** at entry plus **MTM margin** for daily losses. If your balance falls below the maintenance level you get a margin call.

## Leverage by numbers (illustrative)

1 Nifty lot = 24,000 × 75 = **₹18 lakh** exposure. At roughly 10% margin you post ₹1.8 lakh. A 1% index move = ₹18,000 = **10% of your margin**.

# Forwards, futures, options compared

| Feature | Forward | Future | Option |
|---|---|---|---|
| Venue | OTC | Exchange | Exchange or OTC |
| Terms | Customised | Standardised | Standardised |
| Obligation | Both sides | Both sides | Buyer: right; seller: obligation |
| Upfront | None | Margin | Premium (buyer) |
| Settlement | At maturity | Daily mark-to-market | At expiry or exercise |
| Counterparty risk | High | Very low (clearing house) | Very low on exchange |
| Buyer max loss | Unlimited | Unlimited | Premium only |

### Forward

Bank sells you USD 1 lakh in 3 months at ₹88.90. Customised, no cash today.

### Future

Same idea, but traded on an exchange with daily margin settlement.

### Option

Pay a premium for a one-sided right. Non-linear payoff.

# Forward example: farmer and flour mill

Farmer expects **100 tonnes of wheat** at harvest. Mill fears price rises. They agree a forward at **₹25,000/tonne**.

| Spot at harvest | Farmer result | Mill result |
|---|---|---|
| ₹22,000 | **+₹3 lakh** vs market | **−₹3 lakh** |
| ₹25,000 | ₹0 | ₹0 |
| ₹28,000 | **−₹3 lakh** (forgone) | **+₹3 lakh** |

- Zero-sum: one side gains exactly what the other loses relative to spot.
- Both gain **certainty**, which is the real purpose.
- **Default risk:** at ₹28,000 the farmer is tempted to sell elsewhere. Exchanges fix this with margin and daily settlement.

# Forward deep dive: a currency forward for an importer

An Indian manufacturer imports a machine priced at **USD 500,000**, payable in 3 months. Spot ₹88.00; the bank quotes a 3-month forward at **₹89.00**.

| Spot in 3 months | Cost with no hedge | Cost with forward | Forward vs spot |
|---|---|---|---|
| ₹91.00 | ₹4.55 cr | ₹4.45 cr | **Saves ₹10 lakh** |
| ₹89.00 | ₹4.45 cr | ₹4.45 cr | ₹0 |
| ₹87.00 | ₹4.35 cr | ₹4.45 cr | **Costs ₹10 lakh extra** |

## Why do it?

The firm has quoted its customer in rupees. Fixing the import cost protects its margin.

## Forward points

89.00 − 88.00 = ₹1.00. It reflects the interest-rate gap between India and the US, not a forecast.

## Features in action

Customised amount and date, no upfront premium, settled with the bank at maturity. Early exit needs the bank to agree.

# Forward pricing: the cost-of-carry idea

A forward or future price is set by **no-arbitrage**, not by opinion: `F = S × (1 + r × T) − dividends`. Think of it as a conservation law: no free lunch.

| Input | Value (illustrative) |
|---|---|
| Spot S | ₹1,400 |
| Interest rate r | 6.5% a year |
| Time T | 1 month = 1/12 year |
| Fair futures price | 1,400 × (1 + 0.065/12) = **₹1,407.58** |
| If a ₹10 dividend is due this month | 1,407.58 − 10 = ₹1,397.58 |

### Future trades at ₹1,415 (too high)

Borrow, buy the stock at 1,400, sell the future at 1,415. At expiry deliver and repay 1,407.58: riskless profit about **₹7.42/share**. Arbitrageurs push the price down.

### Future trades at ₹1,400 (too low)

Short the stock, invest the proceeds, buy the future. Gain about ₹7.58/share. Real costs keep small gaps alive.

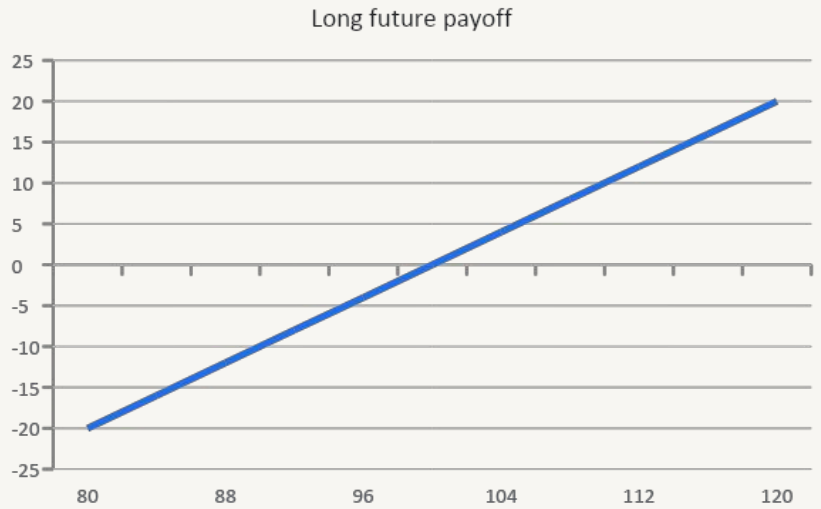
# Futures worked example: linear payoff

## Nifty future (illustrative)

Buy 1 lot at 24,000. Lot size 75.

| Close | Profit |
|---|---|
| 24,100 | +₹7,500 |
| 24,000 | ₹0 |
| 23,900 | **−₹7,500** |

Profit = (Close − 24,000) × 75




**Insight:** forward/future payoffs are straight lines with unlimited loss on either side. Options bend that line.

# Futures: mark-to-market day by day

Long 1 Nifty future at 24,000, lot size 75. The exchange settles profit or loss **every day**.

| Day | Settlement price | Change | Day P/L | Cumulative |
|---|---|---|---|---|
| 1 | 24,100 | +100 | **+₹7,500** | +₹7,500 |
| 2 | 24,040 | −60 | **−₹4,500** | +₹3,000 |
| 3 | 23,900 | −140 | **−₹10,500** | −₹7,500 |
| 4 | 24,000 | +100 | **+₹7,500** | ₹0 |

Day 3: ₹10,500 leaves your account that day. If the balance drops below maintenance margin you must top up or the position is closed. Final P/L equals what a single settlement at 24,000 would give: **zero**.

# Futures deep dive: index vs stock, basis and rollover

| | Index futures | Stock futures |
|---|---|---|
| Underlying | Nifty 50, Bank Nifty | Single shares such as Reliance |
| Settlement | Cash (rupee difference) | Physical delivery of shares in India |
| Used for | Portfolio hedge, market view | Stock-specific view or hedge |
| Lot size | Set by exchange (Nifty example: 75) | Set by exchange per stock |

## Basis

Basis = futures − spot. Example: Nifty spot 24,000, near-month future 24,080, basis 80. It **shrinks to zero** at expiry.

## Rollover

To keep a position, close the near month and open the next. Sell near at 24,060, buy next at 24,130: roll cost 70 × 75 = **₹5,250**.

## Expiry rules

Dates are set by the exchange and change over time. Always check current rules before trading.

# Hedging a portfolio with index futures

A fund manager holds a **₹10 crore** portfolio that tracks the Nifty. Fearing a dip, she **sells Nifty futures** instead of selling stocks.

| Step | Calculation |
|---|---|
| Contract value | 24,000 × 75 = ₹18 lakh |
| Lots to sell | portfolio ÷ contract = 10 cr ÷ 18 lakh ≈ **56 lots** |

| Nifty moves | Portfolio P/L | 56 short futures P/L | Net |
|---|---|---|---|
| −5% (22,800) | **−₹50 lakh** | **+₹50.4 lakh** | ≈ ₹0 |
| +5% (25,200) | **+₹50 lakh** | **−₹50.4 lakh** | ≈ ₹0 |

**Insight:** a futures hedge removes risk in *both* directions. Use put options to keep the upside.

# Same trade, two instruments: gold forward vs gold future

A jeweller needs **1 kg of gold** in 3 months. Illustrative price ₹1,00,000 per 10 g; she locks ₹1,01,000 per 10 g (₹1.01 crore).

| Feature | Forward with a bank | Gold future on an exchange |
|---|---|---|
| Contract | Custom weight and date | Standard lot and expiry |
| Counterparty | The bank (credit risk) | Clearing house |
| Cash flows | Settled at maturity | Daily mark-to-market; margin about ₹5 lakh |
| Exit early | Negotiate with the bank | Reverse the trade any time |
| Price rises to ₹1,10,000 | Gain ₹9 lakh at maturity | Same ₹9 lakh, collected daily |
| Price falls to ₹95,000 | Loss ₹6 lakh at maturity | Same ₹6 lakh, paid daily via margin calls |

# Swaps: turning floating interest into fixed

A **swap** exchanges cash flows on a notional amount. A company has a ₹100 crore loan at **MIBOR + 1%** and fears rising rates. It enters an interest rate swap: pays **7% fixed**, receives MIBOR.

| MIBOR | Loan interest | Swap net | Total cost |
|---|---|---|---|
| 6% | ₹7 cr (6+1) | pay 7%, get 6% = −₹1 cr | ₹8 cr = **8%** |
| 8% | ₹9 cr (8+1) | pay 7%, get 8% = +₹1 cr | ₹8 cr = **8%** |
| 5% | ₹6 cr (5+1) | pay 7%, get 5% = −₹2 cr | ₹8 cr = **8%** |

### Result

Floating cost becomes a fixed 8%. Budget certainty, but no gain if rates fall.

### Other swaps

**Currency swaps** for foreign loans; **commodity swaps** for airlines locking fuel cost.

### Note

Swaps are OTC, so counterparty risk matters. Only the net interest is paid; the notional is not exchanged.

# Meaning of an option

An option gives its buyer the **right but not the obligation** to buy or sell an underlying at strike **K** on or before expiry, in return for a **premium**.

## Buyer (long)

- Pays premium
- Chooses whether to exercise
- Loss limited to premium

## Seller / writer (short)

- Receives premium
- Must honour if exercised
- Margin required, risk can be large

## Think of it as...

A diode: no current below a threshold, a linear response beyond it. Or a deductible-style insurance policy.

# Options in everyday life: ESOPs and insurance

## An ESOP is a call option

Employee gets **1,000 options**, exercise price ₹500, vesting in 2 years.

| Share price | Value |
|---|---|
| ₹900 | **₹4,00,000** |
| ₹500 | ₹0 |
| ₹450 | ₹0, not exercised |

Payoff = max(S − 500, 0) × 1,000. Downside is zero.

## Car insurance is a put option

Car insured for ₹8 lakh, premium ₹20,000 a year.

| Car value after accident | Claim |
|---|---|
| ₹0 (total loss) | ₹8,00,000 |
| ₹5 lakh | About ₹3,00,000 |
| No accident | ₹0, premium lost |

A small premium buys protection against a large loss, the logic of a put.

# The seller side: writing options

Sellers (writers) receive the premium but carry the **obligation**. Example: sell 1 lot of Nifty 24,500 call at premium 100 (lot 75). Premium received: **₹7,500**.

| Nifty at expiry | Call value | Writer P/L per unit | On 75 |
|---|---|---|---|
| 24,300 | 0 | +100 | **+₹7,500** |
| 24,600 | 100 | 0 | ₹0 (break-even) |
| 24,800 | 300 | −200 | **−₹15,000** |
| 25,500 | 1,000 | −900 | **−₹67,500** |

**Asymmetry:** gain is capped at the premium; loss grows without limit. This is why writers face margin and why a naked short call is dangerous.

# Call and put options

| | Call | Put |
|---|---|---|
| Right to | Buy at K | Sell at K |
| Buyer expects | Price to rise | Price to fall |
| Value at expiry | max(S − K, 0) | max(K − S, 0) |
| In-the-money (ITM) | S > K | S < K |
| At-the-money (ATM) | S ≈ K | S ≈ K |
| Out-of-the-money (OTM) | S < K | S > K |
| Analogy | Token advance on a flat | Car insurance |

**Activity (3 min):** K=100, premium 5, spot at expiry 112. Call buyer profit = 12 − 5 = ₹7. Put buyer profit = 0 − 5 = −₹5.

# Moneyness on a sample option chain

Nifty spot = 24,000 (illustrative premiums).

| Strike | Call premium | Call status | Intrinsic + time | Put premium | Put status |
|---|---|---|---|---|---|
| 23,800 | 290 | ITM | 200 + 90 | 60 | OTM |
| 24,000 | 150 | ATM | 0 + 150 | 140 | ATM |
| 24,200 | 70 | OTM | 0 + 70 | 280 | ITM |

- ATM options carry the most time value; deep ITM options are mostly intrinsic.
- OTM options are cheap but need a large move, so they usually expire worthless.
- Put premium at 24,200 = 200 intrinsic + 80 time value.

# American vs European; what is in a premium?

## European

Exercise **only at expiry**. NSE index options (Nifty, Bank Nifty) are European style.

## American

Exercise **any time** up to expiry. Common for US stock options. Not about geography.

## Premium

**Intrinsic value** (what it is worth if exercised now) + **time value** (chance of moving further).

**Factors:** spot vs strike, time to expiry, volatility, interest rates. Time value decays to zero at expiry (*theta decay*), which favours sellers.

# What moves the premium: the Greeks in plain words

## Delta

Premium change per ₹1 move in the underlying. ATM call about 0.5, so Nifty +100 raises it by about 50.

## Theta

Daily time decay. A call losing ₹4/day hurts buyers and helps sellers. Decay speeds up near expiry.

## Vega

Sensitivity to volatility. Uncertain markets (elections, results) raise all premiums.

## Rho

Sensitivity to interest rates. Minor for short-dated options.

**Worked example:** ATM call premium 150, delta 0.5. Nifty rises 100 points → premium ≈ 200. If nothing moves for 5 days at theta −4 → premium ≈ 130.

# Black-Scholes: the core idea

Black and Scholes (1973), with Merton, showed that a European option can be priced by **copying its payoff** with the underlying plus borrowing. By no-arbitrage, the option must cost what the copy costs. (Scholes and Merton shared the 1997 Nobel Prize in Economics.)

## Replication

Hold $\Delta$ units of the index, financed by borrowing. Rebalance as the price moves so the random part cancels and the position is risk-free.

## Risk-neutral pricing

Since risk is hedged away, value = discounted expected payoff, with the index growing at $r - q$.

## The surprise

The expected market return does **not** appear in the formula. Only volatility matters.

## Our SPX 7,700 call as a replicating portfolio

Hold 0.5375 index units (worth **4,120.7**), financed by borrowing **3,846.4**. Net cost = 4,120.7 − 3,846.4 = **274.4**, which is the Black-Scholes price. The two terms of the formula are exactly these two pieces.

# Where the model comes from: random walks and lognormal prices

The model treats the index as a random walk with drift: $dS/S = \mu\,dt + \sigma\,dW$. Each small step is a percentage change plus a random normal shock.

| Term | Meaning |
|---|---|
| μ dt | Average drift (replaced by r − q when pricing) |
| σ dW | Random shock; dW is normal with variance dt |
| Result | S at expiry is **lognormal**: always positive, returns are normal |

## The √T rule

Variance grows linearly with time, so the spread of outcomes grows with √T, like diffusion of heat or noise accumulating in a signal.

## Our example

σ = 16%, T = 101/365: σ√T = 0.0842. One-standard-deviation range for the index about **7,048 to 8,340** (from 7,666.45). Over a full year: about 6,533 to 8,997.

# Assumptions 1-3: price process, volatility, continuity

| Assumption | What it means | Why the model needs it | Reality and consequence |
|---|---|---|---|
| 1. Lognormal (geometric Brownian motion) | Returns are normally distributed; price cannot go negative | Gives a closed-form answer through N(·) | Real returns have fat tails and skew; far out-of-the-money options are mispriced |
| 2. Constant volatility | One σ for the whole life of the option and all strikes | Single parameter to estimate | Volatility changes (VIX moves) and differs by strike: the volatility smile. Fixes: stochastic-volatility models such as Heston |
| 3. Continuous price paths | No jumps or gaps | Allows a perfect, continuous hedge | Overnight gaps, earnings, crashes break the hedge. Fix: jump-diffusion models (Merton 1976) |

# Assumptions 4-7: rates, frictionless trading, exercise style

| Assumption | What it means | Reality and consequence |
|---|---|---|
| 4. Constant r and known dividend yield q | Discounting and index growth use fixed rates | Rates and dividends change; minor for short options, larger for long-dated (LEAPS) ones |
| 5. Frictionless markets | No fees or taxes; borrow and lend at r; short selling allowed; trade continuously | Bid-ask spreads, margin and discrete rebalancing create hedging error; dealers charge for it |
| 6. No arbitrage | No riskless profit available | Broadly true; the model price is a no-arbitrage reference point |
| 7. European exercise | Exercise only at expiry | American options can be exercised early (dividends, deep in-the-money puts); use binomial trees or approximations. **SPX options are European**, so Black-Scholes fits them directly |

# Stress-testing the assumptions: three real-world failures

## 1. Crash of Oct 19, 1987

The S&P 500 fell about 20% in one day. With 16% annual volatility, daily $\sigma \approx 1.0\%$, so the fall was more than **20 standard deviations**: impossible under lognormal prices. Since then, markets charge extra for out-of-the-money puts.

## 2. The smile in our own examples

We used **16%** for the 7,700 call and **20%** for the 7,200 put. Same model, different $\sigma$ by strike. Implied volatility absorbs the model error.

## 3. Discrete hedging

Rebalancing once a day, not continuously, leaves hedging P&L noise. This is gamma risk (see the Greeks slides).

# From assumptions to formula: derivation roadmap

- **1. Build the replicating portfolio:** V = Δ·S + B (index units plus cash).
- **2. Remove randomness:** choose Δ = ∂V/∂S so the shock term cancels. The position is now riskless, so it must earn r.
- **3. Get the equation:** $\partial V/\partial t + \tfrac{1}{2}\sigma^2 S^2 \cdot \partial^2 V/\partial S^2 + (r - q)S \cdot \partial V/\partial S - rV = 0$, with terminal condition V = max(S − K, 0) at expiry.
- **4. Equivalent view:** V = e^(−rT) × E[max(ST − K, 0)] under risk-neutral growth r − q.
- **5. Evaluate the expectation:** it splits into two pieces: the expected index received when in-the-money (gives S·e^(−qT)·N(d1)) and the strike times the chance of exercise (gives K·e^(−rT)·N(d2)).

# The formula: every symbol explained

$$C = S \cdot e^{(-qT)} \cdot N(d1) - K \cdot e^{(-rT)} \cdot N(d2) \quad P = K \cdot e^{(-rT)} \cdot N(-d2) - S \cdot e^{(-qT)} \cdot N(-d1)$$

$$d1 = [\ln(S/K) + (r - q + \sigma^2/2) \cdot T] / (\sigma \cdot \sqrt{T}) \quad d2 = d1 - \sigma \cdot \sqrt{T}$$

| Symbol | Meaning | Units | Call example |
|---|---|---|---|
| S | Current index level | points | 7,666.45 |
| K | Strike price | points | 7,700 |
| T | Time to expiry | years (days/365) | 0.2767 |
| r | Risk-free rate (continuous) | per year | 4.5% (assumed) |
| q | Dividend yield | per year | 1.2% (assumed) |
| σ | Volatility: std dev of log returns | per year | 16% (assumed) |
| N(x) | Standard normal cumulative probability | 0 to 1 | N(0.0987) = 0.5393 |
| n(x) | Normal density, $e^{(-x^2/2)}/\sqrt{(2\pi)}$ | — | used in the Greeks |

# Anatomy of d1 and d2

Read d1 as the **distance from the index to the strike, measured in standard deviations**, with a small adjustment for growth.

| Piece of d1 | Call 7,700 | Put 7,200 | What it does |
|---|---|---|---|
| ln(S/K) | −0.0044 | +0.0628 | Moneyness: how far spot is from strike |
| $(r - q + \sigma^2/2)\cdot T$ | 0.0127 | 0.0106 | Expected drift over the option life |
| $\sigma\sqrt{T}$ | 0.0842 | 0.0894 | Scale of uncertainty (the denominator) |
| **d1** | **0.0987** | **0.8203** | Standardised distance |
| **d2 = d1 − σ√T** | **0.0145** | **0.7309** | N(d2) = chance of finishing above K |

- The call is almost at-the-money (d1 near 0), so N(d2) ≈ 51%.
- The put is about 0.8 standard deviations out-of-the-money, so N(−d2) ≈ 23%.
- d1 is larger than d2 because N(d1) also weights by how much the index is worth when exercised.

# Sanity checks: limits and put-call parity

| Situation | What the formula gives | Why |
|---|---|---|
| $T \to 0$ | Price → intrinsic value $\max(S - K, 0)$ | No time left for movement |
| $\sigma \to 0$ | Call → $\max(S \cdot e^{-qT} - K \cdot e^{-rT}, 0)$ | No uncertainty: the option is a forward |
| S very large | Call ≈ $S \cdot e^{-qT} - K \cdot e^{-rT}$; delta → $e^{-qT}$ | Exercise is nearly certain |
| S very small | Call → 0 | Exercise is nearly impossible |
| Any inputs | $C - P = S \cdot e^{-qT} - K \cdot e^{-rT}$ | Put-call parity (no arbitrage) |

**Parity check, strike 7,700, σ 16%:** C = 274.45, P = 238.13, difference 36.32 = 7,641.0 − 7,604.7. ✓

**Rule of thumb:** at-the-money call ≈ $0.4 \times S \times \sigma\sqrt{T} = 0.4 \times 7{,}666 \times 0.0842 \approx 258$, close to the exact 274 (the gap is the drift term).

# CBOE example 1: SPX 7,700 call, Jan 15, 2027

SPX options on CBOE are European, cash-settled, $100 multiplier. S = 7,666.45 (S&P 500 close, Oct 1, 2026). **Assumed:** r 4.5%, q 1.2%, σ 16%. Valuation date Oct 6, 2026, so T = 101/365 = 0.2767.

| Step | Value |
|---|---|
| d1 = [ln(7666.45/7700) + 0.0458 × 0.2767] / (0.16 × 0.5260) | **0.0987** |
| d2 = d1 − 0.0842 | **0.0145** |
| N(d1) and N(d2) | 0.5393 and 0.5058 |
| S·e^(−qT)·N(d1) | ≈ 4,120.7 |
| K·e^(−rT)·N(d2) | ≈ 3,846.4 |
| **Call price C** | **≈ 274.4 points ≈ $27,445 per contract** |

- Risk-neutral chance of finishing above 7,700: **50.6%**
- Break-even at expiry: 7,700 + 274.4 = **7,974** (+4.0% from spot)

# CBOE example 2: SPX 7,200 put, Dec 18, 2026

Same S = 7,666.45; strike about 6% below spot; T = 73/365 = 0.2000. **Assumed:** r 4.5%, q 1.2%, σ **20%** (puts trade at higher volatility: the skew).

| Step | Value |
|---|---|
| d1 = [ln(7666.45/7200) + 0.053 × 0.2] / (0.2 × 0.4472) | **0.8203** |
| d2 = d1 − 0.0894 | **0.7309** |
| N(−d1) and N(−d2) | 0.2060 and 0.2324 |
| K·e^(−rT)·N(−d2) | ≈ 1,658.3 |
| S·e^(−qT)·N(−d1) | ≈ 1,575.5 |
| **Put price P** | **≈ 82.8 points ≈ $8,285 per contract** |

- Chance of finishing below 7,200: about **23%**
- Break-even: 7,200 − 82.8 = **7,117** (−7.2% from spot)
- Insurance cost ≈ 1.1% of spot. If the index fell 8% at once (vol unchanged) the put would be worth ≈ 305, about 3.7× its cost.

# Greeks overview: formulas and meaning

| Greek | Definition | Call formula | Put formula |
|---|---|---|---|
| Delta Δ | ∂V/∂S | e^(−qT)·N(d1) | −e^(−qT)·N(−d1) |
| Gamma Γ | ∂²V/∂S² | e^(−qT)·n(d1) / (S·σ·√T) | same as call |
| Vega | ∂V/∂σ | S·e^(−qT)·n(d1)·√T | same as call |
| Theta Θ | −∂V/∂T | −S·e^(−qT)·n(d1)·σ/(2√T) − r·K·e^(−rT)·N(d2) + q·S·e^(−qT)·N(d1) | −S·e^(−qT)·n(d1)·σ/(2√T) + r·K·e^(−rT)·N(−d2) − q·S·e^(−qT)·N(−d1) |
| Rho ρ | ∂V/∂r | K·T·e^(−rT)·N(d2) | −K·T·e^(−rT)·N(−d2) |

Greeks are **partial derivatives** of the price: each answers "how much does the option move if only this input changes?" Reported units here: points per index point (delta), per 1% volatility (vega), per calendar day (theta), per 1% rate (rho).

# Delta in depth: the hedge ratio

**Delta** = change in option price per 1-point move in the index. Calls: 0 to +1. Puts: −1 to 0. Put delta = call delta − e^(−qT).

| Call (T = 101 days, σ 16%) | Price | Delta |
|---|---|---|
| Strike 7,200 (in-the-money) | 597.8 | 0.812 |
| Strike 7,700 (at-the-money) | 274.4 | 0.538 |
| Strike 8,200 (out-of-the-money) | 96.0 | 0.257 |

| Spot (call K = 7,700) | 7,300 | 7,666 | 8,000 |
|---|---|---|---|
| Delta | 0.313 | 0.538 | 0.725 |

- **Hedge ratio:** 1 SPX call = 0.5375 × 100 = **53.75 index units** of exposure. Long 10 calls ≈ 537.5 units.
- **Rough probability:** delta ≈ chance of finishing in-the-money (N(d1) = 54% vs N(d2) = 51%).
- **Moves with the market:** delta rises as the index rises, which leads to gamma.

# Gamma in depth: the curvature

**Gamma** = rate of change of delta ($\partial\Delta/\partial S$). Same for calls and puts. Highest for at-the-money options near expiry, because delta flips quickly from about 0 to about 1.

| Days to expiry (call K = 7,700) | 101 | 60 | 30 | 7 | 1 |
|---|---|---|---|---|---|
| Price | 274.4 | 201.9 | 134.0 | 54.5 | 12.5 |
| Gamma | 0.0006 | 0.0008 | 0.0011 | 0.0023 | 0.0055 |

## Check with our call

Index +100: delta goes from 0.5375 to **0.5978**. Gamma predicted 0.5375 + 0.00057 × 100 = 0.5945 (close).

## Who owns gamma

**Long options = long gamma:** profit from large moves either way. **Short options = short gamma:** hurt by large moves, paid by time decay.

## By moneyness

Gamma at 101 days: strike 7,200: 0.0004; 7,700: 0.0006; 8,200: 0.0005. It peaks near the money.

# Theta in depth: time decay

**Theta** = change in price per day as expiry approaches, other things equal. It is negative for long calls and puts and accelerates near expiry.

| Days to expiry (call K = 7,700) | 101 | 60 | 30 | 7 | 1 |
|---|---|---|---|---|---|
| Price | 274.4 | 201.9 | 134.0 | 54.5 | 12.5 |
| Theta per day | −1.60 | −1.98 | −2.66 | −5.08 | −11.48 |

## Why it accelerates

Time value scales with $\sqrt{T}$, so the last days lose the most value per day.

## Theta pays for gamma

The model satisfies $\Theta = rV - (r - q)S\Delta - \frac{1}{2}\sigma^2S^2\Gamma$. For our call, $\frac{1}{2}\sigma^2S^2\Gamma/365 \approx$ **1.26** of the 1.60 daily decay: the cost of owning gamma.

## Buyer vs seller

Buyers pay theta each day. Sellers earn it but carry gamma and vega risk.

# Vega in depth: sensitivity to volatility

**Vega** = change in price per 1 percentage-point rise in volatility. Positive for all long options. Not a Greek letter; it is just a name.

| Call K = 7,700 | Spot 7,300 | Spot 7,666 | Spot 8,000 |
|---|---|---|---|
| Vega (per 1% vol) | 13.59 | 15.96 | 13.94 |

| Days to expiry | 101 | 60 | 30 | 7 | 1 |
|---|---|---|---|---|---|
| Vega | 15.96 | 12.36 | 8.76 | 4.18 | 1.41 |

- Peaks at-the-money; grows with $\sqrt{T}$, so long-dated options are the most volatility-sensitive.
- **Example:** if implied volatility rises from 16% to 19.5% the call gains about 3.5 × 15.96 ≈ **55.9** points (exact repricing: 55.6).
- Volatility is not constant in reality, so vega risk is real. The VIX index summarises 30-day implied volatility from SPX option prices.
- The 7,200 put has lower vega (9.75) because it is further out-of-the-money.

# Rho, and the Greeks cheat sheet

**Rho** = change in price per 1 percentage-point rise in interest rates. Higher rates lift forward prices, which helps calls and hurts puts.

| Option | Rho per +1% rate |
|---|---|
| Call 7,700 (101 days) | +10.64 |
| Put 7,700 (101 days) | −10.40 |
| Put 7,200 (73 days) | −3.32 |

| Greek | Long call | Long put | Largest when |
|---|---|---|---|
| Delta | + | − | Deep in-the-money |
| Gamma | + | + | At-the-money, near expiry |
| Theta | − | − | At-the-money, near expiry |
| Vega | + | + | At-the-money, long maturity |
| Rho | + | − | Long maturity |

Rho is small for short-dated options, so it matters mainly for long-dated ones. Short positions flip every sign.

# Greeks in action: explaining one day of P&L

Long the SPX 7,700 call (price 274.45). Next day: index **+1%** (7,666.45 → 7,743.11, +76.66), implied vol **−1 point** (16% → 15%), **1 day** passes.

| Source | Formula | Points |
|---|---|---|
| Delta | 0.5375 × 76.66 | **+41.21** |
| Gamma | ½ × 0.00057 × 76.66² | **+1.80** |
| Vega | 15.96 × (−1) | **−15.96** |
| Theta | 1 day | **−1.60** |
| **Sum of Greeks** | | **+25.45** |
| **Full repricing** | 300.09 − 274.45 | **+25.64** |

**Insight:** the Greeks explain almost the whole move (residual 0.19 from higher-order terms). A good rise in the index was largely offset by a fall in volatility, which is how traders diagnose "why did my option not gain more?"

# Delta hedging in practice: a worked example

A dealer **sells 10 SPX 7,700 calls** at 274.45 and collects 10 × 100 × 274.45 = **$274,450**. The position is short about 537.5 index units of delta.

| Step | Action | Numbers |
|---|---|---|
| 1. Hedge | Buy index exposure equal to delta (e.g. via futures or an ETF) | Buy ≈ 537.5 units (≈ $4.12 million) |
| 2. Index +100 | Short calls lose ≈ $56,600; hedge gains $53,750 | **Net ≈ −$2,850 (gamma loss)** |
| 3. Rebalance | Delta is now 0.598, so hold 597.8 units | Buy ≈ 60 more |
| 4. Time passes | Theta earned: 10 × 100 × 1.60 | **≈ +$1,600 per day** |

**Break-even move:** gamma loss equals theta gain when $\frac{1}{2}\Gamma \cdot \Delta S^2 = |\Theta|$, so $\Delta S \approx \sqrt{(2 \times 1.60 / 0.00057)}$ **≈ 75 points ≈ 1% of the index**, about one daily standard deviation at 16% volatility. If the index moves more than implied volatility says, the short-option dealer loses.

# Implied volatility, skew and model limits

## Implied volatility

Traders quote options in volatility. Solve for the σ that matches the market price. If the 7,700 call traded at 330, implied vol would be **19.5%** (vs 16% assumed).

## Skew

Out-of-the-money puts trade at higher implied vol than calls. One σ for all strikes is the model weakness.

## Other limits

- American options need trees
- Jumps and fat tails
- VIX options use Black-76
- Rates and dividends are not constant

# Which derivative for which need?

| Situation | Tool | Why |
|---|---|---|
| Farmer wants a fixed harvest price | Forward or commodity future | Price certainty |
| Importer must pay USD in 3 months | Currency forward, or USD call | Forward fixes cost; call keeps the benefit if rupee strengthens |
| Fund manager fears a market dip | Short index futures or put | Futures remove both ways; put keeps upside |
| Airline worried about fuel cost | Futures or commodity swap | Locks input cost |
| Company with floating-rate loan | Interest rate swap | Fixed payments |
| Trader wants a limited-loss bet | Buy a call or put | Loss capped at premium |
| Employee wants a floor on share value | Protective put | Downside protection |

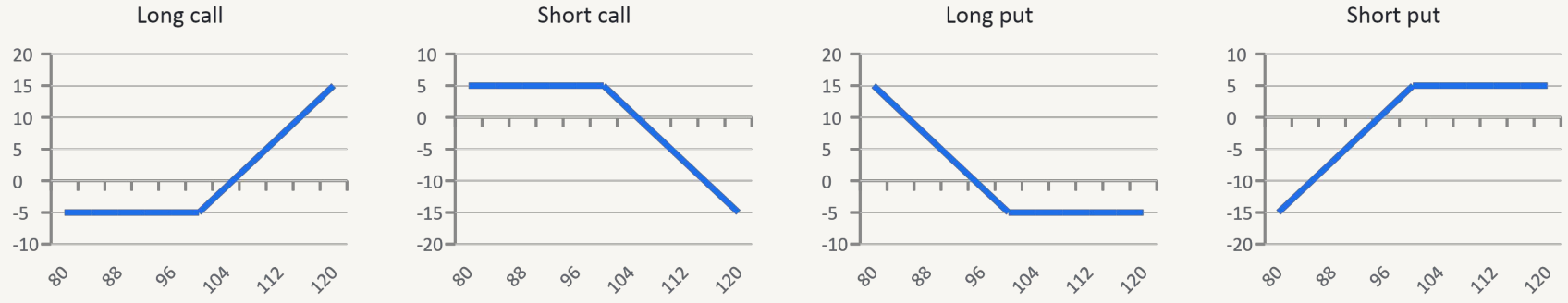
# Practice cases (group activity)

In pairs: name the **risk**, the **tool** and the **reason**.

| # | Situation |
|---|---|
| 1 | A steel maker must buy 500 tonnes of coking coal in 2 months and fears a price jump. |
| 2 | A software firm will receive EUR 2 million in 90 days and fears the euro falling against the rupee. |
| 3 | You hold ₹5 lakh of shares, fear a fall for one month, but want to keep any gains. |
| 4 | A startup has a floating-rate loan and interest rates are rising. |
| 5 | A trader expects Nifty to rise but can afford to lose at most ₹10,000. |
| 6 | A farmer wants a fixed sale price from a local trader he trusts. |

# Call and put payoff diagrams

Strike 100, premium 5. Profit per share at expiry.




| Spot | 80 | 90 | 95 | 100 | 105 | 110 | 120 |
|---|---|---|---|---|---|---|---|
| Long call | −5 | −5 | −5 | −5 | 0 | +5 | +15 |
| Long put | +15 | +5 | 0 | −5 | −5 | −5 | −5 |

# Payoff summary and motives

| Position | Max profit | Max loss | Break-even | Motive |
|---|---|---|---|---|
| Long call | Unlimited | ₹5 | 105 | Bullish, limited risk |
| Short call | ₹5 | Unlimited | 105 | Mildly bearish, income |
| Long put | ₹95 | ₹5 | 95 | Bearish or insurance |
| Short put | ₹5 | ₹95 | 95 | Mildly bullish, buy cheaper |

## Leverage example

Stock ₹100, call ₹5. At ₹120: share +20%, call (15/5) **+300%**. At ₹100: share 0%, call **−100%**.

## Risk warning

Naked short calls have unlimited risk. Beginners should prefer defined-risk positions.

# Hedging: concept

**Hedging** = taking a derivative position that offsets an existing risk. It reduces uncertainty; it does not maximise profit. Think factor of safety.

### Forward hedge

Locks a price. Certainty, but upside is surrendered.

### Option hedge

Locks a worst case and keeps upside. Costs a premium.

### Portfolio hedge

Buy an OTM Nifty put to cushion a diversified equity portfolio.

# Protective put on an equity portfolio

Investor holds a Nifty-like portfolio worth **₹18 lakh** (24,000 × 75). Buys a **23,500 put at 150** (cost ₹11,250).

| Nifty at expiry | Unhedged P/L | Put payoff | Hedged net |
|---|---|---|---|
| 26,000 | +₹1,50,000 | 0 | **+₹1,38,750** |
| 24,000 | 0 | 0 | **−₹11,250** |
| 23,500 | −₹37,500 | 0 | −₹48,750 |
| 22,000 | **−₹1,50,000** | +₹1,12,500 | **−₹48,750** |

**Max loss** = (24,000 − 23,500 + 150) × 75 = **₹48,750**, however far the market falls. The premium is a deductible, like insurance.

# Limits of hedging: cost, basis and over-hedging

## Cost drag

Rolling puts every month steadily erodes returns if markets do not fall.

## Basis risk

Hedging a single stock portfolio with Nifty puts leaves a gap if the stock moves differently from the index.

## Hedge ratio

Hedge only the exposure you are sure about.

## Cautionary case (illustrative)

Exporter sells USD 1M forward at 88.90, but the order is cancelled. Rupee weakens to 92. The forward now loses (92 − 88.90) × 1M = **₹31 lakh** with no receivable to offset it. The hedge became a speculation.

# IT exporter: the problem

A Bengaluru IT services firm will receive **USD 1,000,000** in 3 months. Costs are in rupees.

## Exposure

If the rupee **strengthens** (USD falls), rupee revenue shrinks.

## Illustrative market data

- Spot ₹88.00 → ₹8.80 cr
- 3-month forward ₹88.90
- USD put (INR call): strike ₹88.50, premium ₹0.60 (₹6 lakh total)

## Choices

- Do nothing
- Lock with a forward
- Buy a put option

# IT exporter: outcomes

| Spot in 3 months | No hedge | Forward 88.90 | Put (net of premium) |
|---|---|---|---|
| 85.00 | **₹8.50 cr** | ₹8.89 cr | ₹8.79 cr (floor) |
| 88.90 | ₹8.89 cr | ₹8.89 cr | ₹8.79 cr |
| 92.00 | ₹9.20 cr | ₹8.89 cr | **₹9.14 cr** |

### No hedge

Revenue swings ₹8.5 to ₹9.2 cr. Budgeting is hard.

### Forward

₹8.89 cr certain. Gives up gains if the rupee weakens.

### Option

Floor of ₹8.79 cr, upside retained. Premium is the insurance cost.

**Group activity (4 min):** A firm imports machinery priced in USD, payable in 3 months. What is its risk and which hedge fits? *(Rupee weakening; buy forward or a USD call.)*

# Covered call: concept

### Structure

Own the stock **+ sell a call** on it. The shares "cover" the delivery obligation.

### When

Neutral to mildly bullish; want income from shares you plan to hold.

### Trade-off

Premium income and a small cushion, in exchange for **capped upside**.

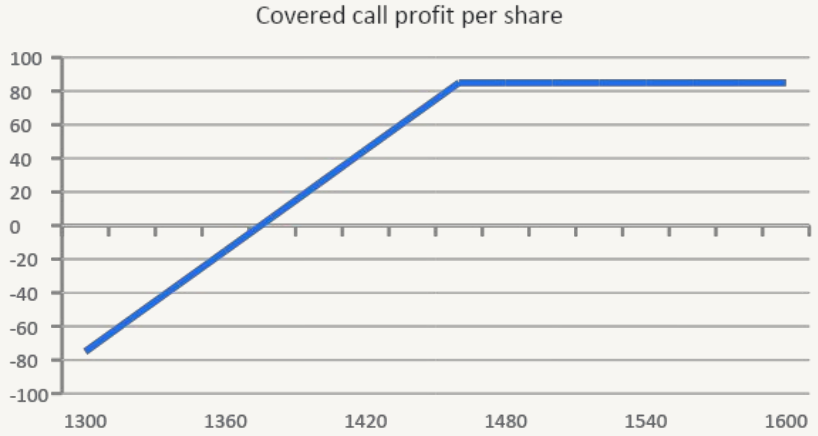
**Engineering view:** a clipper circuit. You cut off the top of the waveform in return for steady income.

# Covered call lifecycle: strike, expiry, assignment, rolling

- **Step 1:** hold 500 shares at ₹1,400.
- **Step 2:** pick a strike above spot (₹1,460, about 4% OTM) and a one-month expiry.
- **Step 3:** sell the call and collect ₹25 × 500 = ₹12,500 immediately.
- **Step 4a:** expiry below ₹1,460 → call lapses, keep shares and premium. Yield about 25/1,400 = **1.8% for the month** (not guaranteed to repeat).
- **Step 4b:** expiry above ₹1,460 → shares are sold at ₹1,460 (assignment).
- **Rolling:** before expiry, buy back the call and sell a later or higher-strike one to stay invested or give up less upside.

# Covered call on Reliance shares

Investor holds **500 shares** bought at ₹1,400 and sells one call: strike ₹1,460, premium ₹25. Premium received: **₹12,500**. (Illustrative.)

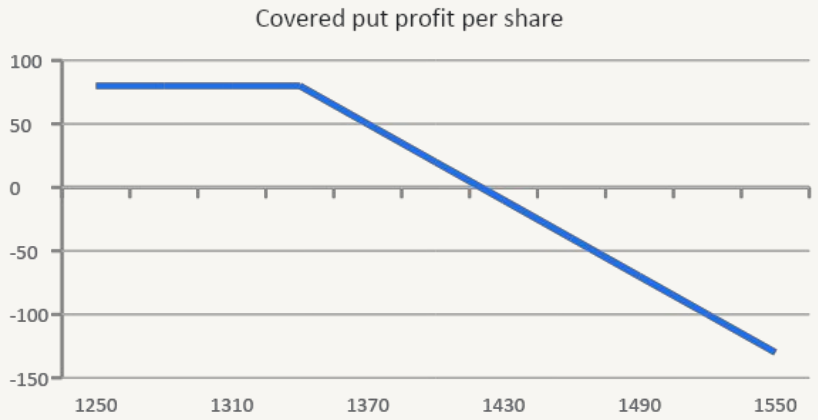



## Outcomes at expiry

| Price | Net/share | On 500 |
|---|---|---|
| 1,350 | −25 | −₹12,500 |
| 1,375 | 0 | break-even |
| 1,400 | +25 | +₹12,500 |
| 1,460 | +85 | +₹42,500 (max) |
| 1,600 | +85 | +₹42,500 |

- **Max profit** = (1,460 − 1,400) + 25 = ₹85/share
- **Break-even** = 1,400 − 25 = ₹1,375
- **Opportunity cost:** a rally to ₹1,600 still earns only ₹85; shares are called away at 1,460

# Covered put and comparison




## Example (short at 1,400, sell 1,340 put for 20)

- Max profit = 60 + 20 = **₹80**
- Break-even = 1,420
- Max loss: unlimited on a rally

| | Covered call | Covered put |
|---|---|---|
| View | Neutral to mildly bullish | Neutral to mildly bearish |
| Positions | Long stock + short call | Short stock + short put |
| Max loss | Large (stock to zero) | Unlimited |

# Choosing a strategy: view, risk and reward

| Market view | Strategy | Risk | Reward |
|---|---|---|---|
| Strongly bullish | Long call | Premium | Unlimited |
| Mildly bullish | Short put / covered call | Large downside | Premium (+ capped gain) |
| Neutral | Covered call | Stock falls | Premium |
| Mildly bearish | Covered put | Unlimited rally loss | Premium + capped gain |
| Strongly bearish | Long put | Premium | Large |
| Protect holdings | Protective put / forward | Premium / lost upside | Loss floor |

# Risk management and common mistakes

## Common mistakes

- Selling naked options for small premiums
- Ignoring lot size, so exposure is bigger than expected
- Forgetting time decay as a buyer
- Over-leveraging with margin
- No exit plan on expiry day

## Checklist before any trade

- Define maximum loss first
- Size so one loss is survivable
- Know break-even and payoff shape
- Check margin needed
- Write down the exit rule

SEBI studies have reported that a large majority of individual F&O traders lose money. Treat derivatives as risk-management tools first.

# Check your understanding

**Q1. Which contract is marked to market daily?**
A) Forward  B) Future  C) Neither

**Q2. Call strike 200, premium 8, spot 215 at expiry. Profit?**
A) ₹15  B) ₹7  C) ₹8

**Q3. Break-even of a long put, K=500, premium 20?**
A) ₹520  B) ₹500  C) ₹480

**Q4. Which options can be exercised only at expiry?**
A) American  B) European  C) Neither

**Q5. Why might the exporter prefer the put to the forward?**
A) It costs nothing  B) Keeps upside if rupee weakens  C) No margin

**Q6. Covered call bought at ₹800, premium ₹15. Break-even?**
A) ₹785  B) ₹815  C) ₹800

**Q7. Main drawback of a covered call?**
A) Unlimited loss  B) Capped upside  C) No income

**Q8. A short put has which max loss?**
A) Premium  B) Unlimited  C) K − premium

# Key Takeaways

- Derivatives derive value from an underlying and serve hedging, speculation and arbitrage.
- Forwards/futures = obligations (linear). Options = rights (non-linear).
- Call = right to buy; put = right to sell. Buyers risk only the premium.
- Hedging reduces uncertainty: forwards lock price, options lock a worst case.
- Covered call = income plus capped upside.

**The slides are for educational purposes only; These are not investment advice. Prices are illustrative.**